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ELECENG 3CL4 (Introduction to Control Systems)
Final Examination - April 2025

Department of Electrical and Computer Engineering, McMaster University
Instructors: Dr. Tim Davidson (C01) and Dr. Shahrukh Athar (C02), Winter 2025
Friday, April 25, 2025, 12:30 PM to 3:00 PM (150 minutes)

Examination Conditions This test is conducted under standard examination conditions. In particular:

- You are not to communicate in any manner with any other student.
- You are not to look at the questions nor start writing until told to do so by an invigilator.
- When the invigilator announces that the test is over, you are to stop writing *immediately*.

The penalties for violating these or any other standard examination conditions will be severe, including a zero on the test and the commencement of formal disciplinary proceedings.

Instructions

- There are five (5) questions on this examination paper. Answer all five questions in the space provided in this exam paper. You may use the back of the pages if necessary.
- **When given permission to start writing, you must ensure that your copy of the exam paper is complete and bring any discrepancies to the attention of your exam invigilator.**
- **Write your name and student number (nine-digit McMaster ID) on the front page of this examination paper. Encircle your section at the top of this page.**
- Write your answers neatly and in a logical fashion. We may refuse to mark answers which are difficult to decipher.
- Direct answers without proper justifications will lead to a reduced grade. However, justifications should be concise and to-the-point.
- You may write your test in pencil, but booklets written in pencil or with pages torn out will not be re-graded.
- **Closed-books, closed-notes.**
- **Some useful information is provided towards the end of this exam paper.**
- **Aids permitted:**
McMaster Standard Calculator (Casio FX-991MS/MS+).
- Total Marks: 100 (mark allocations for questions/parts are shown within questions).

Q1 (20)	Q2 (20)	Q3 (20)	Q4 (20)	Q5 (20)	Total (100)

Question 1 (20 marks) This question is composed of four short parts (a-d). **Answers are expected to be short and concise.** We have provided more space than is necessary. Answer the following questions:

a) (6 marks) A plant with input $x(t)$ and output $y(t)$ is described by the differential equation

$$\frac{dy(t)}{dt} + ay(t) = \frac{dx(t)}{dt} + bx(t).$$

If $x(t)$ is a unit step function, the steady-state value of $y(t)$ is $\lim_{t \rightarrow \infty} y(t) = 1/4$.

i) • (4 marks) Determine the ratio $\frac{b}{a}$.

ii) • (2 marks) Describe how the step response depends upon the value of a .

$$a) \quad x(t) = \frac{A}{s}$$

$$f_v = \frac{1}{4}$$

$$y'(t) + ay(t) = x'(t) + bx(t)$$

i) $sY(s) + aY(s) = sX(s) + bX(s)$ ii) $\therefore$ 1st order system, step response has decaying exp. term
 $\therefore a$ is reciprocal for time constant term

$$Y(s)(s+a) = X(s)(s+b)$$

$$G(s) = \frac{Y(s)}{X(s)} = \frac{s+b}{s+a}$$

$$\therefore f_v = \frac{b}{a}$$

$\hookrightarrow$ type 0 & step input:

$\therefore$ if a large, then f_v small & system respond quickly
 if a small, then f_v large & system respond slowly

$$e_{ss} = f_v$$

$$= \lim_{s \rightarrow 0} s \cdot G(s) \cdot \frac{1}{s}$$

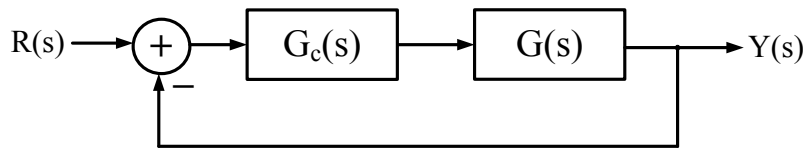
$$= \lim_{s \rightarrow 0} G(s)$$

$$= \lim_{s \rightarrow 0} \frac{s+b}{s+a}$$

$$= \lim_{s \rightarrow 0} \frac{0+b}{0+a}$$

$$\frac{1}{4} = \frac{b}{a}$$

b) (4 marks)

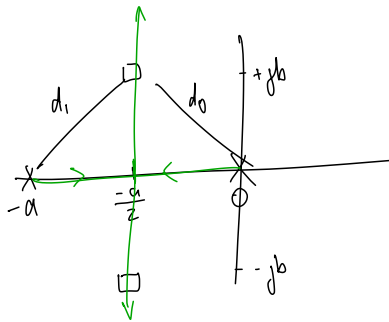


Consider the closed-loop feedback control system above, where $G(s) = \frac{1}{s(s+a)}$ and $G_c(s) = K$.

Derive an expression for the value of K , as a function of $b > 0$, that puts the closed-loop poles at the point $s = -a/2 \pm jb$.

(Hint: The magnitude criterion for a particular value of s to be a closed-loop pole may simplify your analysis.)

Cl. poles @ $s = \frac{-a}{2} \pm jb$



$$G_c(s) G(s) = \frac{K}{s(s+a)}$$

↳ p's @ 0, -a

↳ no z

$$K = d_0 \cdot d_1$$

$$= \sqrt{\left(\frac{a}{2}\right)^2 + b^2} \cdot \sqrt{\left(\frac{a}{2}\right)^2 + b^2}$$

$$= \left(\frac{a}{2}\right)^2 + b^2$$

$$K = \frac{a^2}{4} + b^2$$

c) (4 marks) A plant has a transfer function $G(s) = \frac{3}{s+2}$. In the steady state, what is the magnitude and phase of the sinusoidal signal at the output of the system if the sinusoid at the input of the system has magnitude 1, phase 0, and

i) • (2 marks) the frequency is 2 radians-per-second.

ii) • (1 mark) the frequency approaches zero from above.

iii) • (1 mark) the frequency is very large.

$$G(s) = \frac{3}{s+2}$$

• output sinus. mag & phase = ?

• input sinus. mag = 1 & phase = 0

i) $\omega = 2 \frac{\text{rad}}{\text{s}}$

$$G(s) = \frac{3}{s+2}$$

$$G(j\omega) = \frac{3}{j\omega + 2}$$

$$G(j2) = \frac{3}{j2 + 2}$$

$$= \frac{3}{2 + 2j} \rightarrow \text{denom} = 2 + 2j = \frac{2j + d}{2} \quad \theta = 45^\circ$$

$$= \frac{3}{d \cdot e^{j\theta}}$$

$$= \frac{3}{d} \cdot e^{-j\theta}$$

$$G(j2) = \frac{3}{2\sqrt{2}} \cdot e^{-j45^\circ}$$

$$\text{mag.} = \frac{3}{2\sqrt{2}} \quad \text{phase} = -45^\circ$$

$$d = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

ii) when $\omega \rightarrow 0^+$: $G(j\omega) = \frac{3}{j\omega + 2}$

$$G(j0) = \frac{3}{j0 + 2}$$

$$= \frac{3}{2}$$

$$\text{mag} = \frac{3}{2} \quad \& \quad \text{phase} = 0^\circ$$

iii) when $\omega \rightarrow \infty$: $G(j\omega) = \frac{3}{j\omega + 2}$

$$G(j\infty) = \frac{3}{j\infty + 2}$$

$$|G(j\infty)| = \left| \frac{3}{j\infty + 2} \right|$$

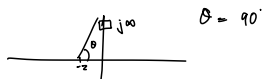
$$= 0$$

$$\angle G(j\omega) = \angle \frac{3}{2 + e^{j90^\circ}}$$

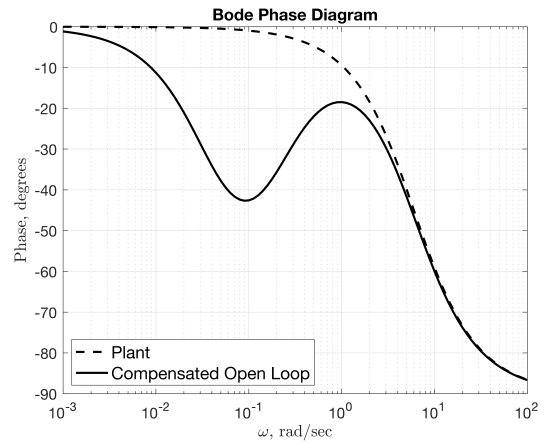
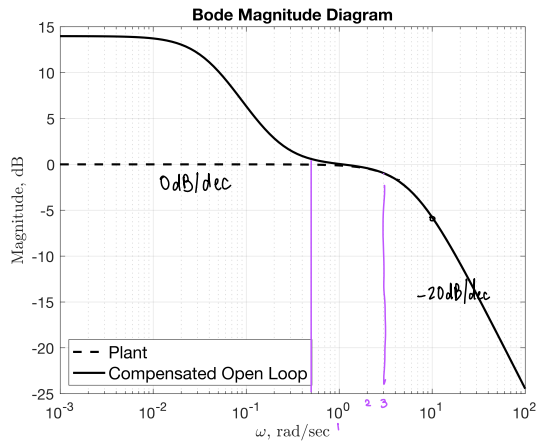
$$= \angle e^{-j90^\circ}$$

$$= -90^\circ$$

$$\text{mag} = 0 \quad \& \quad \text{phase} = -90^\circ$$



- d) (6 marks) The Bode magnitude and phase diagrams of a plant frequency response $G(j\omega)$ (dashed lines) and a compensated open-loop frequency response $G_c(j\omega)G(j\omega)$ (solid lines) are given in the following figures.



- i) • (1 mark) What is the DC gain of the plant?
- ii) • (1 mark) Postulate the structure of the plant transfer function.
- iii) • (2 marks) Qualitatively describe the impact of the compensator.
- iv) • (1 mark) What is the DC gain of the compensated open loop?
- v) • (1 mark) Postulate the structure of the compensator transfer function.

i) $G(0) = ?$
 $= 0 \text{ dB} \quad \& \quad 0$
 $= 10^{\frac{0}{20}}$
 $= 1$

v) $G_c(s) = \frac{s+z}{s+p}$

$20 \log_{10} \left(\frac{z}{p} \right) = 14 \text{ dB}$

$\frac{z}{p} = 10^{\frac{14}{20}}$

$\frac{z}{p} = 5$

$z \doteq 0.5$

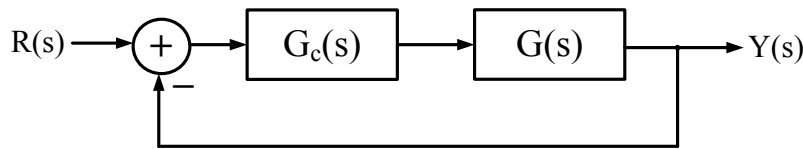
- ii) • $-20 \text{ dB/dec} \rightarrow 1 \text{ pole} \quad \& \quad \text{no zero}$
 • $0 \rightarrow -90^\circ \rightarrow 1 \text{ pole} \quad \& \quad \text{no zero}$
 • $G(0) = 1$

$\hookrightarrow G(s) = \frac{a}{s+a} \quad , \quad a \doteq 3$

- iii) • add low freq gain
 • add $-ve \theta$ @ low freq.
 • no change to high freq.
 • should be lag compensator

iv) $G_c G(0) = 14 \text{ dB}$
 $= 10^{\frac{14}{20}}$
 $= 10^{0.7}$
 $\doteq 5$

Question 2 (20 marks)



Consider the closed-loop feedback control system above, where

$$G(s) = \frac{25}{s(s+10)(s+30)} \quad \text{and} \quad G_c(s) = \frac{K(s+5)}{(s+20)}$$

- (2 marks)** Write the open-loop transfer function $L(s)$. What is the type number of the open-loop system?
- (4 marks)** Derive the closed-loop transfer function $T(s)$ as a ratio of polynomials in s .
- (12 marks)** Sketch the complete Root Locus for this system as K increases from zero to infinity, and show all steps, i.e., your work should show the allowable segments of the real line, the number of asymptotes (if any) along with their associated angles and the centroid, the value of K that might put a pair of closed-loop poles on the $j\omega$ -axis along with the position of the $j\omega$ -axis poles.
- (2 marks)** Would it be correct to say that the closed-loop system is stable for all values of K . Justify your answer based on the Root Locus sketch obtained in Part (c).

$$\begin{aligned} a) \quad L(s) &= G_c(s) G(s) \\ &= \frac{25K(s+5)}{s(s+10)(s+20)(s+30)} \end{aligned}$$

1 p @ origin $\rightarrow$ type 1

$$\begin{aligned} b) \quad T(s) &= \frac{L(s)}{1+L(s)} \\ &= \frac{25K(s+5)}{s(s+10)(s+20)(s+30)} \div \left[\frac{s(s+10)(s+20)(s+30)}{s(s+10)(s+20)(s+30)} + \frac{25K(s+5)}{s(s+10)(s+20)(s+30)} \right] \\ &= \frac{25K(s+5)}{s(s+10)(s+20)(s+30)} \div \frac{25K(s+5) + s(s+10)(s+20)(s+30)}{s(s+10)(s+20)(s+30)} \\ &= \frac{25K(s+5)}{25K(s+5) + s(s+10)(s+20)(s+30)} \\ &= \frac{25K(s+5)}{25Ks + 125K + (s^2 + 10s)(s^2 + 50s + 600)} \\ &= \frac{25Ks + 125K}{s^4 + 60s^3 + 1100s^2 + (25K + 6000)s + 125K} \end{aligned}$$

[More space for your answer to Question 2.]

c) $1 + G_c(s) G(s) = 1 + k P(s)$, $k = 25k$

$$P(s) = \frac{(s+5)}{s(s+10)(s+20)(s+30)}$$

$$\# p = n = 4$$

$$\# z = m = 1$$

$$\# \text{ asymptotes} = n - m = 4 - 1 = 3$$

$$\text{angle} = (2l+1) \frac{180}{n-m}$$

$$= (2l+1) \frac{180}{3}$$

$$= (2l+1) 60^\circ$$

$$= 60^\circ, 180^\circ, 300^\circ$$

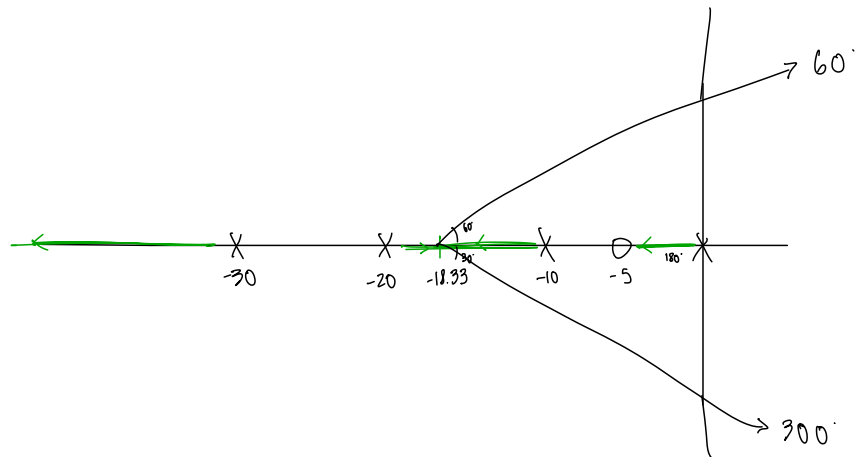
$$\sigma_A = \frac{\sum p - \sum z}{n-m}$$

$$= \frac{(0 - 10 - 20 - 30) - (-5)}{3}$$

$$= \frac{-60 + 5}{3}$$

$$= \frac{-55}{3}$$

$$= -18.33$$



$k = ?$

$$q(s) = 0 = s^4 + 60s^3 + 1100s^2 + (25k + 6000)s + 125k$$

row	$4 = s^4$			
	1	1100	125k	
	$3 = s^3$	60	$25k + 6000$	0
	$2 = s^2$	b_1	b_0	0
	$1 = s^1$	c_1	0	
	$0 = s^0$	d_1	0	

$$b_1 = \frac{-1}{60} \begin{vmatrix} 1 & 1100 \\ 60 & 25k + 6000 \end{vmatrix}$$

$$= \frac{-1}{60} (25k + 6000 - 66000)$$

$$= \frac{-1}{60} (25k - 60000)$$

$$= \frac{60000 - 25k}{60}$$

$$b_0 = \frac{-1}{60} \begin{vmatrix} 1 & 125k \\ 60 & 0 \end{vmatrix}$$

$$= \frac{-1}{60} (0 - 60 \cdot 125k)$$

$$= 125k$$

$$c_1 = \frac{-1}{b_1} \begin{vmatrix} 60 & 25k + 6000 \\ b_1 & b_0 \end{vmatrix}$$

$$= \frac{-1}{b_1} (60b_0 - b_1(25k + 6000))$$

$$= 25k + 6000 - \frac{60b_0}{b_1}$$

$$d_1 = \frac{-1}{c_1} \begin{vmatrix} b_1 & b_0 \\ c_1 & 0 \end{vmatrix}$$

$$= \frac{-1}{c_1} (0 - b_0 c_1)$$

$$= b_0$$

$$= 125k$$

[More space for your answer to Question 2.]

* force row 1 = zero row
row 2 = aux row

$$0 = c_1 s'$$

$$= -625k^2 + 900,000k + 36,000,000$$

$$= k^2 - 1440k - 576,000$$

$$k = 1766.135, -326.135$$

$$0 = a(s)$$

$$= b_1 s^2 + b_0$$

$$0 = \frac{60000 - 25k}{60} s^2 + 125k, \quad k = 1766.135$$

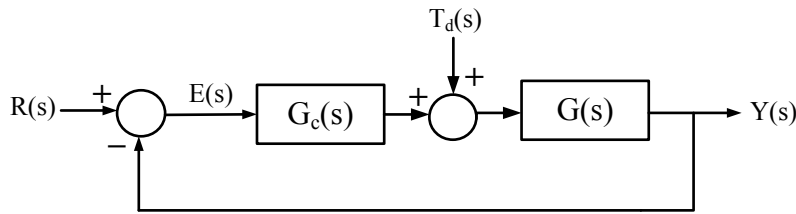
$$s^2 = -835.89$$

$$s = \pm j 28.91$$

$$\therefore k = 1766.135 \quad \text{for} \quad s_{1,2} = \pm j 28.91$$

d) not stable for $k > 1766.135$

Question 3 (20 marks)



Consider the closed-loop feedback control system above, where $G(s) = \frac{200}{s(s-7)}$

- a) (14 marks) Design a compensator, $G_c(s)$, so that the closed-loop system achieves the following:
- It is stable.
 - A percentage overshoot of **12.314%** for a step input.
 - A 2% settling time of **0.25 sec** for a step input.
- b) (4 marks) Check whether the compensator designed in Part (a) attenuates low-frequency disturbances by a factor of at least 35 dB. In case it does not, modify your compensator so that this additional design requirement is also met.
- c) (2 marks) Comment on whether the final compensator designed in Part (b) will meet the transient design requirements as specified in Part (a). Justify your answer, i.e., explain why or why not deviations will occur.

a) $PO = 12.314\%$ for step input
 $T_s = 0.25 \text{ s}$

$$G(s) = \frac{200}{s(s-7)}$$

$$\zeta = \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}}$$

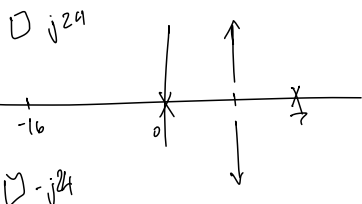
$$= 0.5547$$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$\omega_n = \frac{4}{\zeta T_s} = 28.844$$

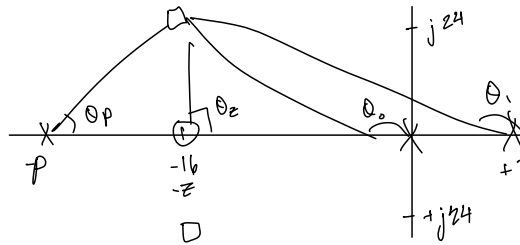
$$s_{1,2} = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$$

$$= -16 \pm j24$$



$\therefore$ prop not gonna work
 $\therefore$ try lead

$$G_c(s) = \frac{k_c(s+z)}{(s+p)}$$



$$180^\circ = \text{all } \theta_z - \text{all } \theta_p$$

$$= \theta_z - \theta_p - \theta_1 - \theta_0$$

$$\theta_p = \theta_z - \theta_1 - \theta_0 - 180^\circ$$

$$= 90^\circ - \left(180 - \tan^{-1}\left(\frac{24}{16}\right)\right) - \left(180 - \tan^{-1}\left(\frac{24}{16+7}\right)\right) - 180^\circ$$

$$= 90^\circ - 123.7^\circ - 133.8^\circ - 180^\circ$$

$$= -347.47^\circ + 360^\circ$$

$$= 12.53^\circ$$

$$\tan \theta_p = \frac{24}{\text{adj}}$$

$$\text{adj} = \frac{24}{\tan \theta_p} = 8$$

$$\text{adj} = 108$$

$$-p = -16 - \text{adj} = -16 - 108$$

$$p = 124$$

[More space for your answer to Question 3.]

$$G_c G = \frac{K_c K_0 (s+16)}{s(s-7)(s+124)}, \quad K = K_c K_0$$

$P(s) = \text{the rest}$

mag. cond.

$$1 = KP(s_0)$$

$$K = \frac{\prod d_p \text{ to } s_0}{\prod d_z \text{ to } s_0}$$

$$K = \frac{d_0 d_1 d_p}{d_z}$$

$\therefore$ use trig

$$K = \frac{28.8 \times 33.2 \times 110.6}{24}$$

$$\approx 4420$$

$$K = K_c K_0$$

$$K_c = \frac{K}{K_0}$$

$$= \frac{4420}{200}$$

$$\approx 22.1$$

$$G_c(s) = 22.1 \frac{(s+16)}{(s+124)}$$

[More space for your answer to Question 3.]

$$b) \mathcal{E}(s) = \frac{-G(s)}{1 + G_c(s)G(s)} T_d(s)$$

$$f_{attenuation} = \frac{G(s)}{1 + G_c(s)G(s)}$$

$$\text{low freq. att} = -35 \text{ dB} = 10^{\frac{-35}{20}} = 0.0178$$

$$\lim_{s \rightarrow 0} |f_{att}| \leq \text{low freq att.}$$

$$\lim_{s \rightarrow 0} \left| \frac{G(s)}{1 + G_c(s)G(s)} \right| \leq 0.0178$$

$$f_{\text{low freq att.}} = 0.0178$$

$\therefore$ type 1 $\rightarrow G(0)$ large

$$\therefore \lim_{s \rightarrow 0} \left| \frac{\cancel{G(s)}}{1 + G_c(s)\cancel{G(s)}} \right|$$

$$= \lim_{s \rightarrow 0} \left| \frac{1}{G_c(s)} \right|$$

$$= \frac{1}{|G_c(0)|}$$

$$= \frac{1}{22.1 \times 16}$$

$$= 0.351 > 0.0178$$

$\therefore$ need lead lag

$$\alpha_{\text{lag}} = \frac{\text{current att.}}{\text{req att.}}$$

$$= \frac{0.351}{0.0178}$$

$$= 19.72 = \frac{z_{\text{lag}}}{p_{\text{lag}}}$$

$$G_{c, \text{lag}}(s) = \frac{s + z_{\text{lag}}}{s + p_{\text{lag}}}, \quad z_{\text{lag}} = 0.1$$

$$p_{\text{lag}} = \frac{z_{\text{lag}}}{\alpha_{\text{lag}}} = \frac{0.1}{19.72} = 0.0051$$

$$\therefore G_c(s) = 22.1 \times \frac{s+16}{s+124} \times \frac{s+0.1}{s+0.0051}$$

C)

• significantly deviate from design reqs because

$\hookrightarrow$ additional zero equivalent to step response derivative

$\hookrightarrow$ but zero close to pole

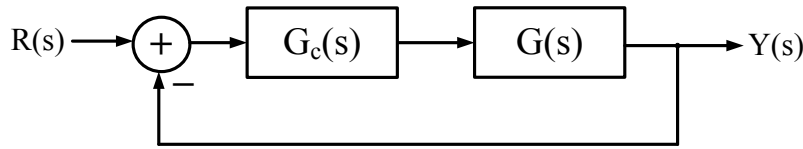
$\hookrightarrow$ but z_{lead} far from p_{lead} , & will reach when $k \rightarrow \infty$, so derivative effect won't be suppressed enough

$\hookrightarrow$ so transient response leads to

$$PO > 12.314\%$$

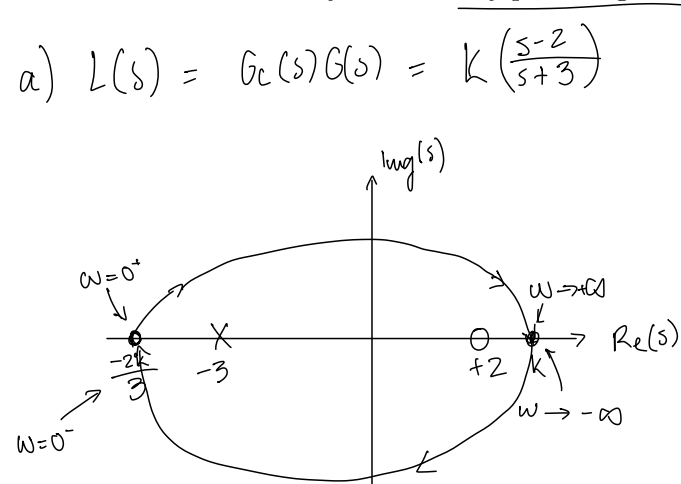
[More space for your answer to Question 3.]

Question 4 (20 marks)



Consider the closed-loop feedback control system above, with $G_c(s) = K$ and $G(s) = \frac{s-2}{s+3}$.

- (12 marks)** Sketch the Nyquist diagram of the open-loop transfer function $KG(s)$. (Note that the features of this sketch will depend on K .)
- (5 marks)** From that diagram, determine the range of values of K for which the closed loop will be stable. Provide a clear explanation of why you believe that this is the appropriate range of values.
- (3 marks)** Using root locus techniques, determine the range of values of K for which the closed loop is stable. Provide a clear explanation of why your root locus analysis either confirms or contradicts the analysis of the Nyquist diagram.

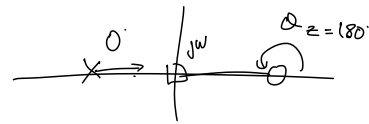


$$L(j\omega) = K \left(\frac{j\omega - 2}{j\omega + 3} \right)$$

$$\omega = 0 : L(j0) = K \left(\frac{j0 - 2}{j0 + 3} \right)$$

$$= \frac{-2K}{3}$$

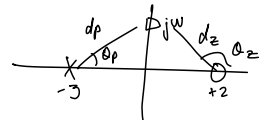
$$\angle L(j0) = \theta_z - \theta_p = 180^\circ$$



$$\omega > 0 : L(j\omega) = K \left(\frac{j\omega - 2}{j\omega + 3} \right)$$

$$|L(j\omega)| = \frac{K d_z}{d_p} = \frac{K \infty}{\infty} = K$$

$$\angle L(j\omega) = \theta_z - \theta_p = 90^\circ - 90^\circ = 0$$



$$\therefore L(-j\omega) = L^*(j\omega)$$

- b) $\therefore$ no O.L poles in RHP
 $\therefore$ stable if no CW encircles around -1

$\hookrightarrow$ true:

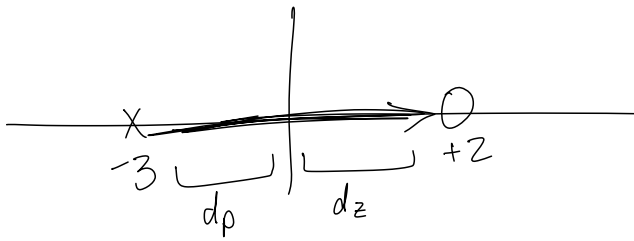
$$\hookrightarrow \text{if } \frac{-2K}{3} > -1$$

$$-2K > -3$$

$$\therefore K < \frac{3}{2}$$

[More space for your answer to Question 4.]

c)



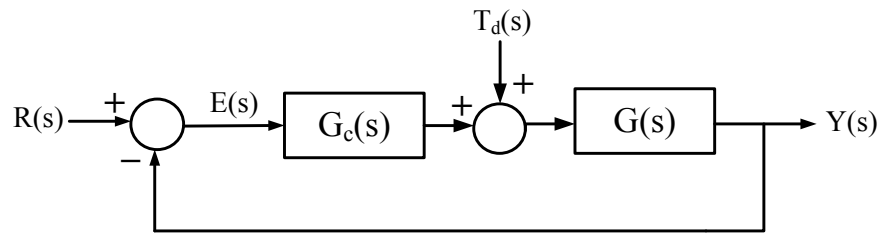
$$\begin{aligned}k &= \frac{d_p}{d_z} \\&= \frac{3}{2} \\&= 1.5\end{aligned}$$

RH gets 1.5 as well

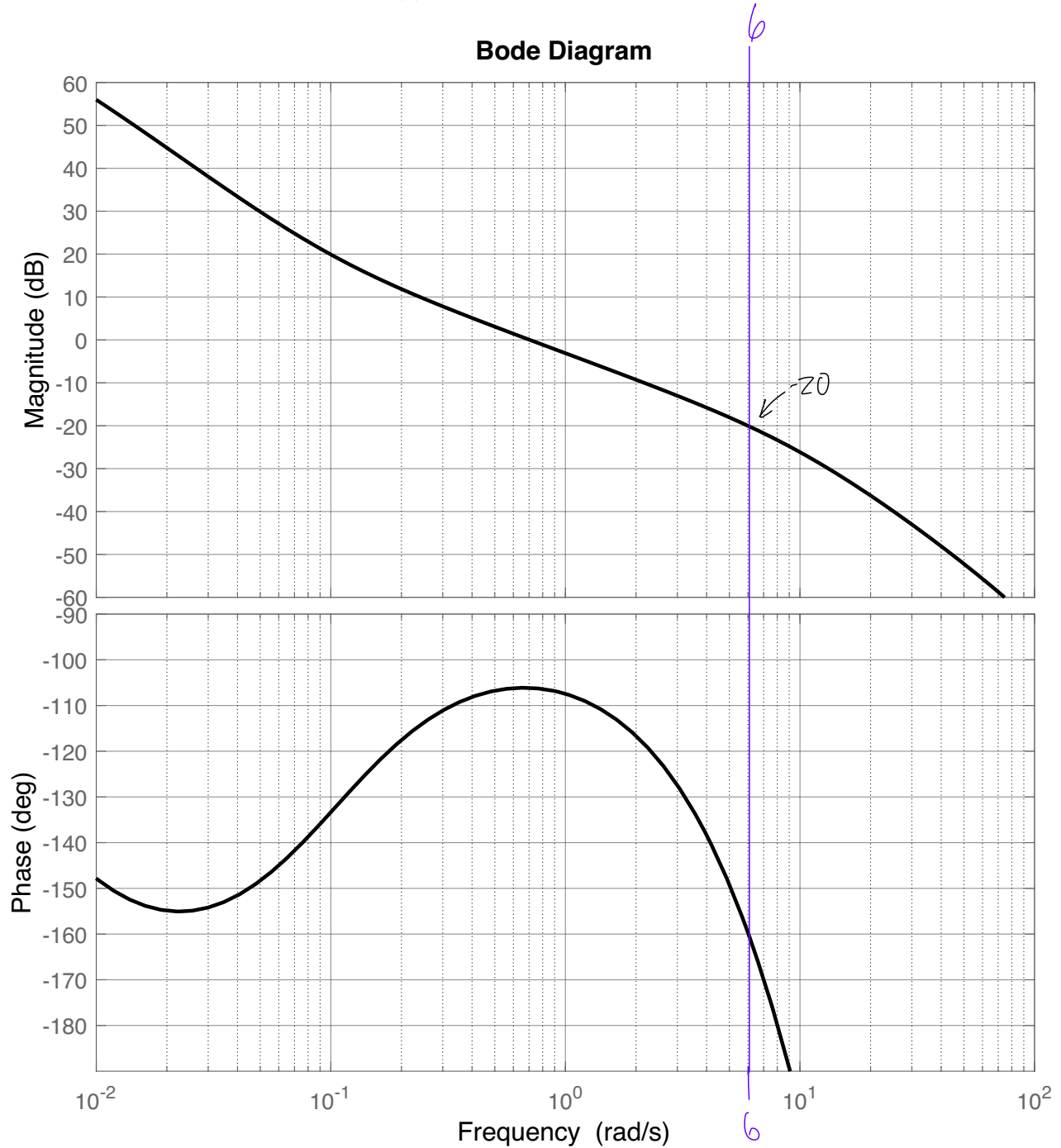
↳ ∴ confirms Nyquist diagram

[More space for your answer to Question 4.]

Question 5 (20 marks) Consider the following closed-loop feedback control system:



The frequency response of the plant, $G(s)$, is given in the following diagram:



a) (14 marks) Design a compensator, $G_c(s)$, that achieves both of the following requirements:

- A percentage overshoot of 18.118% in the closed-loop response to a step input.
- An open-loop crossover frequency of $\omega_c = 6 \text{ rad/sec}$.

b) (4 marks) Check whether the compensator designed in Part (a) reduces the steady-state error to step disturbances to 20% of the step disturbance height. If it does not, then modify your compensator so that this additional design requirement is also met.

c) (2 marks) What is the expected 2% settling time of the compensated system?

$$\bullet PO = 18.118\%$$

$$\bullet \omega_c = 6 \frac{\text{rad}}{\text{sec}}$$

$$\begin{aligned} \bullet \zeta &= f(PO) \\ &= \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}} \\ &= 0.4777 \end{aligned}$$

$$\begin{aligned} \phi_{pm} &= f(\zeta) \\ &= \tan^{-1} \left(\frac{2}{\sqrt{4 + \frac{1}{\zeta^4}} - 2} \right) \\ &= 50^\circ \end{aligned}$$

$$\omega_c = 6 \text{ rad/s}$$

$$\angle G(j\omega) |_{\omega=6} = -160^\circ$$

$$\begin{aligned} \phi_{pm, \text{prop}} &= 180 + \angle G(j\omega) \\ &= 180 - 160 \\ &= 20^\circ \end{aligned}$$

$\therefore \omega_c$ & ϕ_{pm} constraints
 $\therefore$ need lead comp.

$$\text{lead comp: } G_{c, \text{lead}}(s) = \frac{k_c (s+z)}{s+p}$$

$$k_c = ? \text{ for } |L(j\omega_c)| = 1$$

$$\begin{aligned} \phi_{pm, \text{add}} &= \phi_{pm, \text{req}} - \phi_{pm, \text{prop}} \\ &= 50^\circ - 20^\circ \end{aligned}$$

$$= 30^\circ$$

$$\begin{aligned} \alpha_{\text{lead}} &= \frac{1 + \sin(\phi_{pm, \text{add}})}{1 - \sin(\phi_{pm, \text{add}})} \\ &= \frac{1 + \sin(30^\circ)}{1 - \sin(30^\circ)} \end{aligned}$$

$$= 3$$

$$\alpha_{\text{lead}} = \frac{p}{z}$$

$$3 = \frac{p}{z}$$

$$p = 3z$$

$$\omega_c = \sqrt{zp}$$

$$= \sqrt{z \cdot 3z}$$

$$= \sqrt{3z^2}$$

$$6 = \sqrt{3} z$$

$$z = \frac{6}{\sqrt{3}}$$

$$= 3.46$$

$$p = 3z$$

$$= 3(3.46)$$

$$= 10.38$$

$$G_{c, \text{lead}}(s) = \frac{k_c (s+3.46)}{(s+10.38)}$$

$$1 = |G_c(j\omega_c) G(j\omega_c)|$$

$$= \left| k_c \frac{j6+z}{j6+p} \right| |G(j6)|$$

$$= \frac{k_c |j6+z| |G(j6)|}{|j6+p|}$$

$$= \frac{k_c |j6+3.46| |-20 \text{ dB}|}{|j6+10.38|}$$

$$\begin{aligned} x &= 10^{-\frac{20}{20}} = 10^{-1} = 0.1 \\ \sqrt{6^2 + 3.46^2} &= 6.9 \\ \sqrt{6^2 + 10.38^2} &= 11.9 \end{aligned}$$

$$1 = \frac{k_c \times 6.9 \times 0.1}{11.9}$$

$$k_c = \frac{11.9}{6.9 \times 0.1}$$

$$= 17.31$$

$$\therefore G_{c, \text{lead}}(s) = \frac{17.31 (s+3.46)}{(s+10.38)}$$

[More space for your answer to Question 5.]

$$b) \quad \varepsilon(s) = -f_{att}(s) T_d(s) \quad , \quad f_{att}(s) = \frac{G(s)}{1 + G_c(s)G(s)}$$

$$f_{att, req, low \text{ freq}} = \lim_{s \rightarrow 0} |f_{att}(s)|$$

$$20\% = \lim_{s \rightarrow 0} \left| \frac{G(s)}{1 + G_c(s)G(s)} \right|$$

$$0.2 = \left| \frac{\cancel{G(0)}}{1 + G_c(0)\cancel{G(0)}} \right|$$

$$= \frac{1}{G_c(0)}$$

$$0.2 = \frac{1}{\frac{17.31(3.46)}{10.38}}$$

$$0.2 > 0.1733$$

does meet req.

$\therefore$ no need for lag comp.

$$c) \quad T(s) = \frac{4 f_c(\zeta)}{\omega_c} \quad , \quad f_c(\zeta) = \sqrt{4 + \frac{1}{\zeta^4}} - 2$$

$$f_c(0.4777) = 1.678$$

$$= \frac{4 \times 1.678}{6}$$

$$= 1.12 \text{ s}$$

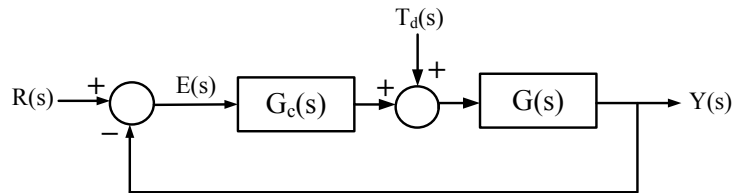
[More space for your answer to Question 5.]

Useful Information

- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

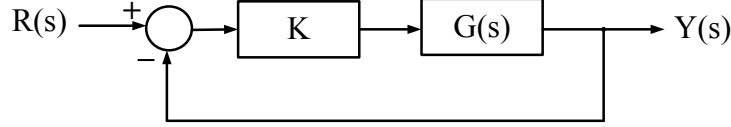
$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} & t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 e^{-at} u(t) &\longleftrightarrow \frac{1}{s+a} & \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \sin(\omega t) u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} & \frac{d^2 f(t)}{dt^2} &\longleftrightarrow s^2 F(s) - sf(0^-) - f'(0^-) \\
 \cos(\omega t) u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} & \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s} \\
 e^{-at} \cos(\omega t) u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t) u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2}
 \end{aligned}$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^M$ and the set of poles is $\{-p_j\}_{j=1}^n$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, when $0 < \zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:
 - $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \theta)$, where $\theta = \cos^{-1}(\zeta)$.
 - 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
 - Percentage (maximum) overshoot calculated from step response, $P.O. = 100 \left[\frac{M_{pt} - fv}{fv} \right]$, where M_{pt} is the peak value and fv is the steady-state value of the step response.
 - Percentage (maximum) overshoot (given ζ), $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$.
 - Peak time, $T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$.
- Consider a feedback control system of the following form:



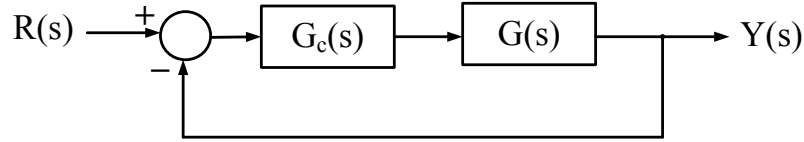
The Laplace Transform of the error is $E(s) = \frac{1}{1+G_c(s)G(s)} R(s) - \frac{G(s)}{1+G_c(s)G(s)} T_d(s)$

- Steady state error constants:
 - Position error constant: $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
 - Velocity error constant: $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
- Consider a feedback control system of the following form:



where $K > 0$ and $G(s)$ takes the form $G(s) = \frac{K_G \prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, with $K_G > 0$.

- The point s_0 is on the root locus if $\sum_{i=1}^M \angle(s_0 + z_i) - \sum_{j=1}^n \angle(s_0 + p_j) = 180^\circ + k360^\circ$ for some integer k .
- The value of the gain K such that s_0 is a pole of the closed loop is $K = \frac{\prod_{j=1}^n |s_0 + p_j|}{K_G \prod_{i=1}^M |s_0 + z_i|}$.
- Consider the following closed-loop system (where the open-loop transfer function is $L(s) = G_c(s)G(s)$):



- The gain margin is $K_{\text{gm}} = \frac{1}{|L(j\omega_x)|} = \frac{1}{|G_c(j\omega_x)G(j\omega_x)|}$, where ω_x is the frequency at which $\angle G_c(j\omega_x)G(j\omega_x) = -180^\circ$.
- K_{gm} (in dBs) = $-20 \log_{10}(|L(j\omega_x)|)$
- The phase margin is $\phi_{\text{pm}} = 180^\circ + \angle L(j\omega_c) = 180^\circ + \angle G_c(j\omega_c)G(j\omega_c)$, where the crossover frequency ω_c is the frequency at which $|G_c(j\omega_c)G(j\omega_c)| = 1$.
- In the frequency domain, for a closed-loop system of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, the 3dB bandwidth is:
 - $\omega_{\text{cl,3dB}} = \omega_n (\sqrt{2 - 4\zeta^2 + 4\zeta^4} + 1 - 2\zeta^2)^{1/2}$
- In the frequency domain, for an open-loop system of the form $L(s) = G_c(s)G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)}$:
 - The phase margin is: $\phi_{\text{pm}} = \tan^{-1}\left(\frac{2}{\sqrt{(4+1/\zeta^4)^{1/2}-2}}\right)$. If ϕ_{pm} is measured in degrees, and if $\zeta \leq \frac{1}{\sqrt{2}}$, then $\phi_{\text{pm}} \approx 100\zeta$.
 - The crossover frequency is: $\omega_c = \omega_n (\sqrt{1 + 4\zeta^4} - 2\zeta^2)^{1/2}$.
 - The 2% settling time is: $T_s = \frac{4f_c(\zeta)}{\omega_c}$, where $f_c(\zeta) = (\sqrt{4 + 1/\zeta^4} - 2)^{1/2}$.
- A first order lead compensator has the form $G_c(s) = \frac{K_c(s+z)}{s+p}$ with $|z| < |p|$ and $\alpha_{\text{lead}} = \frac{p}{z}$. The peak of the phase response of $G_c(j\omega)$ occurs at $\omega = \omega_m = \sqrt{zp}$.
- A first order lag compensator has the form $G_c(s) = \frac{K_c(s+z)}{s+p}$ with $|p| < |z|$ and $\alpha_{\text{lag}} = \frac{z}{p}$.

End of Examination